Passages - The Road to Reality

Plato made it clear that the mathematical propositions the things that could be regarded as unassailably true referred not to actual physical objects (like the approximate squares, triangles, circles, spheres, and cubes that might be constructed from marks in the sand, or from wood or stone) but to certain idealized entities. He envisaged that these ideal entities inhabited a different world, distinct from the physical world. The Platonic world of mathematical forms.

The mathematical forms of Plato's world clearly do not have the same kind of existence as do ordinary physical objects such as tables and chairs. They do not have spatial locations; nor do they exist in time. Objective mathematical notions must be thought of as timeless entities and are not to be regarded as being conjured into existence at the moment that they are first humanly perceived but already 'in existence' since the beginning of time, in the potential timeless sense that they would necessarily be revealed precisely in the form that we perceive them today, no matter at what time or in what location some perceiving being might have chosen to examine them.

Thus, according to Fig. 1.3, the entire physical world is depicted as being governed according to mathematical laws. On this view, everything in the physical universe is indeed governed in completely precise detail by mathematical principles- perhaps by equations, or perhaps by some future mathematical notions fundamentally different from those which we would today label by the term 'equations'. If this is right, then even our own physical actions would be entirely subject to such ultimate mathematical control, where 'control' might still allow for some random behaviour governed by strict probabilistic principles.

Morality has a profound connection with the mental world, since it is so intimately related to the values assigned by conscious beings and, more importantly, to the very presence of consciousness itself. It is hard to see what morality might mean in the absence of sentient beings.

We need to address the question of separating true from false before we can adequately attempt to apply such understanding to separate good from bad.

Hyperbolic geometry, turns out to have all the properties of Euclidean geometry apart from those which need the parallel postulate.

Escher's picture merely makes use of one particular Euclidean representation of hyperbolic geometry. Hyperbolic geometry itself is a more abstract thing which does not depend upon any particular Euclidean representation.

Saccheri failed in his attempt to find a contradiction. He was therefore not able to obtain a proof of the fifth postulate. But in striving for it he, in effect, found something far greater: a new geometry, different from that of Euclid - hyperbolic geometry.

Things like the natural numbers can be conjured literally out of nothing, merely by employing the abstract notion of 'set.

The laws governing the behaviour of the world, at its tiniest scales, is fundamentally governed by the complex-number system.

In mathematics, it is indeed imperative to be absolutely clear that one's equations make strict and accurate sense. However, it is equally important not to be insensitive to 'things going on behind the scenes' which may ultimately lead to deeper insights.

The remarkable general answer for any given power series is that there is some circle in the complex plane centred at 0, called the circle of convergence, with the property that if the complex number: lies strictly, inside the circle then the series converges for that value of z, whereas if z lies strictly outside the circle then the series diverges for that value of z?